POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year: 2025

Programme: BE Computer Full Marks: 100

Course: Data Science and Analytics(New) Pass Marks: 45

Type: Numerical and Short Notes Solved Time : 3hrs.

TOP 5 MOST FREQUENT NUMERICAL QUESTIONS - DETAILED SOLUTIONS

I'll solve each question exactly as it appears in the PDF, with step-by-step calculations and [+1] tracking for repetitions.

🥇 RANK 1: PEARSON CORRELATION COEFFICIENT [Asked 7 times]

Question from Page 1, Question 2a (First Occurrence) [+1]

Question: Calculate the correlation coefficient for the following data by the help of Pearson's correlation coefficient formula:
X - 10, 13, 15, 17, 19 and Y - 5, 10, 15, 20, 25.

Solution:

Step 1: Organize the Data

Observation	X	Y
1	10	5
2	13	10
3	15	15
4	17	20
5	19	25

Observation	X	Y
Total	**74**	**75**

Step 2: Calculate Means

$n = 5$

Mean of X ($\bar{x}$) = $\Sigma X / n = 74/5 =$ **14.8**

Mean of Y ($\bar{y}$) = $\Sigma Y / n = 75/5 =$ **15**

Step 3: Calculate Required Components

X	Y	$(X - \bar{x})$	$(Y - \bar{y})$	$(X - \bar{x})^2$	$(Y - \bar{y})^2$	$(X - \bar{x})(Y - \bar{y})$
10	5	10 - 14.8 = -4.8	5 - 15 = -10	23.04	100	48
13	10	13 - 14.8 = -1.8	10 - 15 = -5	3.24	25	9
15	15	15 - 14.8 = 0.2	15 - 15 = 0	0.04	0	0
17	20	17 - 14.8 = 2.2	20 - 15 = 5	4.84	25	11
19	25	19 - 14.8 = 4.2	25 - 15 = 10	17.64	100	42
Σ				**48.8**	**250**	**110**

Step 4: Apply Pearson's Formula

$r = \Sigma[(X - \bar{x})(Y - \bar{y})] / \sqrt{[\Sigma(X - \bar{x})^2 \times \Sigma(Y - \bar{y})^2]}$

$r = 110 / \sqrt{(48.8 \times 250)}$

$r = 110 / \sqrt{12,200}$

$r = 110 / 110.45$

$\mathbf{r = 0.996}$

Step 5: Interpretation

- **Strength**: 0.996 indicates a **very strong positive correlation** (close to +1)
- **Direction**: Positive (as X increases, Y increases)
- **Interpretation**: There is an almost perfect linear relationship between X and Y

Same Question from Page 7, Question 2a (Second Occurrence) [+2]

Question: Explain Pearson correlation and how it measures the relationship between two numeric variables? Show it by considering an example.

(Same calculation as above would be shown)

Same Question from Page 9, Question (Third Occurrence) [+3]

Question: Calculate and interpret Pearson's co-relation coefficient.

(Same calculation as above would be shown)

Same Question from Page 12, Question 2a (Fourth Occurrence) [+4]

Question: compute Karl Pearson's correlation coefficient and interpret the result.

X	4	7	8	9	5	3
Y	7	9	5	10	8	5

Solution:

Step 1: Organize Data

n = 6

X	Y	XY	X^2	Y^2
4	7	28	16	49
7	9	63	49	81
8	5	40	64	25

X	Y	XY	X^2	Y^2
9	10	90	81	100
5	8	40	25	64
3	5	15	9	25
$\Sigma X=36$	**$\Sigma Y=44$**	**$\Sigma XY=276$**	**$\Sigma X^2=244$**	**$\Sigma Y^2=344$**

Step 2: Apply Pearson's Formula

$r = [n\Sigma XY - (\Sigma X)(\Sigma Y)] / \sqrt{[n\Sigma X^2 - (\Sigma X)^2][n\Sigma Y^2 - (\Sigma Y)^2]}$

$r = [6(276) - (36)(44)] / \sqrt{[6(244) - (36)^2][6(344) - (44)^2]}$

$r = [1656 - 1584] / \sqrt{[1464 - 1296][2064 - 1936]}$

$r = 72 / \sqrt{168 \times 128}$

$r = 72 / \sqrt{21{,}504}$

$r = 72 / 146.64$

$r = 0.491$

Step 3: Interpretation

- **Strength**: 0.491 indicates a **moderate positive correlation**
- **Direction**: Positive
- **Interpretation**: There is a moderate linear relationship between X and Y

Same Question from Page 19, Question 2b (Fifth Occurrence) [+5]

Question: The following data represents age (X) and annual income (Y in thousands):

X	22	25	30	35	40
Y	**25**	**30**	**45**	**50**	**60**

Compute Pearson correlation coefficient and interpret.

Solution:

Step 1: Calculate Components

$n = 5$
$\Sigma X = 22 + 25 + 30 + 35 + 40 = 152$
$\Sigma Y = 25 + 30 + 45 + 50 + 60 = 210$
$\Sigma XY = (22\times25) + (25\times30) + (30\times45) + (35\times50) + (40\times60)$
$= 550 + 750 + 1350 + 1750 + 2400 = 6800$
$\Sigma X^2 = 484 + 625 + 900 + 1225 + 1600 = 4834$
$\Sigma Y^2 = 625 + 900 + 2025 + 2500 + 3600 = 9650$

Step 2: Apply Formula

$r = [n\Sigma XY - (\Sigma X)(\Sigma Y)] / \sqrt{\{[n\Sigma X^2 - (\Sigma X)^2][n\Sigma Y^2 - (\Sigma Y)^2]\}}$

$r = [5(6800) - (152)(210)] / \sqrt{\{[5(4834) - (152)^2][5(9650) - (210)^2]\}}$

$r = [34000 - 31920] / \sqrt{\{[24170 - 23104][48250 - 44100]\}}$

$r = 2080 / \sqrt{\{1066 \times 4150\}}$

$r = 2080 / \sqrt{4,423,900}$

$r = 2080 / 2103.3$

$r = 0.989$

Step 3: Interpretation

- **Very strong positive correlation** (0.989)
- As age increases, income tends to increase strongly
- Nearly perfect linear relationship

Same Question from Page 21, Question 4 (Sixth Occurrence) [+6]

Question: Calculate Pearson's Correlation Coefficient between Age and Income. Show necessary Intermediate steps.

Employee_ID	Age	Salary
01	30	50000

Employee_ID	Age	Salary
02	25	30000
03	22	25000
04	28	40000
05	20	20000

Solution:

Step 1: Organize Data (Salary in thousands for easier calculation)

Age (X)	Salary (Y in 000s)	XY	X^2	Y^2
30	50	1500	900	2500
25	30	750	625	900
22	25	550	484	625
28	40	1120	784	1600
20	20	400	400	400
ΣX=125	**ΣY=165**	**ΣXY=4320**	**ΣX^2=3193**	**ΣY^2=6025**

n = 5

Step 2: Apply Formula

$r = [n\Sigma XY - (\Sigma X)(\Sigma Y)] / \sqrt{\{[n\Sigma X^2 - (\Sigma X)^2][n\Sigma Y^2 - (\Sigma Y)^2]\}}$

$r = [5(4320) - (125)(165)] / \sqrt{\{[5(3193) - (125)^2][5(6025) - (165)^2]\}}$

$r = [21600 - 20625] / \sqrt{\{[15965 - 15625][30125 - 27225]\}}$

r = 975 / √{340 × 2900}

r = 975 / √986,000

r = 975 / 993

r = 0.982

Step 3: Interpretation

- **Strength**: 0.982 (Very strong positive correlation)
- **Direction**: Positive
- **Interpretation**: There is a very strong positive correlation between age and salary. As age increases, salary tends to increase.

Same Question from Page 23, Question 4 (Seventh Occurrence) [+7]

Question: Explain Pearson's Correlation and types of correlations using suitable examples and visuals.

(This is more theoretical but would include the same calculation example)

🥇 RANK 1: NADARAYA-WATSON KERNEL REGRESSION [Asked 7 times]

Question from Page 2, Question 5a (First Occurrence) [+1]

Question: You are given the following data. Assuming that there is an unknown functional relationship y = f(x), estimate the value of y at x = 2.9 using the Nadaraya-Watson regression estimator. Take bandwidth h = 0.5.

x	**1**	**2**	**3**	**4**
y	**-2.3**	**0.5**	**9.3**	**20.8**

Solution:

Step 1: Recall Nadaraya-Watson Formula

$$\hat{y}(x) = \frac{\sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right) Y_i}{\sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right)}$$

Where K is the kernel function. Since not specified, we'll use the **Gaussian kernel**:

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$$

Step 2: Calculate for x = 2.9, h = 0.5

For each data point, calculate $u = (x - X_i)/h$ and $K(u)$

i	X_i	Y_i	$u = (2.9 - X_i)/0.5$	$K(u) = e^{(-u^2/2)}$	$K(u) \times Y_i$
1	1	-2.3	(2.9-1)/0.5 = 1.9/0.5 = 3.8	e^(-(3.8)²/2) = e^(-14.44/2) = e^(-7.22) = 0.00073	0.00073 × (-2.3) = -0.00168
2	2	0.5	(2.9-2)/0.5 = 0.9/0.5 = 1.8	e^(-(1.8)²/2) = e^(-3.24/2) = e^(-1.62) = 0.198	0.198 × 0.5 = 0.099
3	3	9.3	(2.9-3)/0.5 = -0.1/0.5 = -0.2	e^(-(-0.2)²/2) = e^(-0.04/2) = e^(-0.02) = 0.980	0.980 × 9.3 = 9.114
4	4	20.8	(2.9-4)/0.5 = -1.1/0.5 = -2.2	e^(-(-2.2)²/2) = e^(-4.84/2) = e^(-2.42) = 0.089	0.089 × 20.8 = 1.851

Step 3: Sum the Components

$\Sigma K(u) = 0.00073 + 0.198 + 0.980 + 0.089 = \mathbf{1.26773}$

$\Sigma [K(u) \times Y_i] = -0.00168 + 0.099 + 9.114 + 1.851 = \mathbf{11.06232}$

Step 4: Calculate Estimated Value

$$\hat{y}(2.9) = \frac{11.06232}{1.26773} = 8.726$$

Answer: ŷ(2.9) = 8.73 (rounded to 2 decimal places)

Question from Page 7, Question 4a (Second Occurrence) [+2]

Question: You are given the following data. Assuming that there is an unknown functional relationship y = f(x), estimate the value of y at x = 2.9 using the Nadaraya-Watson regression estimator. Take bandwidth h = 0.5.

x	1	2	3	4
y	**-2.3**	**0.5**	**9.3**	**20.8**

(Identical to first question - same solution as above)

Question from Page 10, Question 4b (Third Occurrence) [+3]

Question: Given a new dataset to demonstrate non-parametric regression:
X = [1,2,3,4,5]
Y = [2,4,1,5,3]
Estimate y at x = 2.5 using the Nadaraya-Watson kernel regression with Gaussian kernel:

$$\hat{y}(x) = \frac{\sum_{i=1}^{n} K_h\,(x - X_i)Y_i}{\sum_{i=1}^{n} K_h\,(x - X_i)}$$

where $K_h(u) = \exp\left(-\frac{u^2}{2h^2}\right)$ and bandwidth $h = 1$

Solution:

Step 1: Calculate for x = 2.5, h = 1

For each data point, calculate $u = (x - X_i)/h$ and $K(u) = \exp(-u^2/2)$

i	**X_i**	**Y_i**	**$u = (2.5 - X_i)/1$**	**u^2**	**K(u) = e^($-u^2/2$)**	**K(u) × Y_i**
1	1	2	1.5	2.25	e^(-2.25/2) = e^(-1.125) = 0.325	0.325 × 2 = 0.65
2	2	4	0.5	0.25	e^(-0.25/2) = e^(-0.125) = 0.882	0.882 × 4 = 3.528
3	3	1	-0.5	0.25	e^(-0.125) = 0.882	0.882 × 1 = 0.882
4	4	5	-1.5	2.25	e^(-1.125) = 0.325	0.325 × 5 = 1.625

i	X_i	Y_i	u = (2.5 - X_i)/1	u²	K(u) = e^(-u²/2)	K(u) × Y_i
5	5	3	-2.5	6.25	e^(-6.25/2) = e^(-3.125) = 0.044	0.044 × 3 = 0.132

Step 2: Sum the Components

Σ K(u) = 0.325 + 0.882 + 0.882 + 0.325 + 0.044 = **2.458**

Σ [K(u) × Y_i] = 0.65 + 3.528 + 0.882 + 1.625 + 0.132 = **6.817**

Step 3: Calculate Estimated Value

$$\hat{y}(2.5) = \frac{6.817}{2.458} = 2.774$$

Answer: ŷ(2.5) = 2.77 (rounded to 2 decimal places)

Question from Page 15, Question 4a (OR option) (Fourth Occurrence) [+4]

Question: You are given the following dataset

x	0.5	1.0	1.5	2.0
y	-2.3	0.5	9.3	20.8

Estimate the value of y at x = 1.3 using the Nadaraya-Watson regression estimator with a bandwidth h = 0.2.

Solution:

Step 1: Calculate for x = 1.3, h = 0.2

Using Gaussian kernel K(u) = e^(-u²/2)

i	X_i	Y_i	u = (1.3 - X_i)/0.2	u²	K(u) = e^(-u²/2)	K(u) × Y_i
1	0.5	-2.3	(1.3-0.5)/0.2 = 0.8/0.2 = 4.0	16	e^(-16/2) = e^(-8) = 0.000335	0.000335 × (-2.3) = -0.00077

i	X_i	Y_i	u = (1.3 - X_i)/0.2	u²	K(u) = e^(-u²/2)	K(u) × Y_i
2	1.0	0.5	(1.3-1.0)/0.2 = 0.3/0.2 = 1.5	2.25	e^(-2.25/2) = e^(-1.125) = 0.325	0.325 × 0.5 = 0.1625
3	1.5	9.3	(1.3-1.5)/0.2 = -0.2/0.2 = -1.0	1.0	e^(-1/2) = e^(-0.5) = 0.607	0.607 × 9.3 = 5.6451
4	2.0	20.8	(1.3-2.0)/0.2 = -0.7/0.2 = -3.5	12.25	e^(-12.25/2) = e^(-6.125) = 0.0022	0.0022 × 20.8 = 0.0458

Step 2: Sum the Components

Σ K(u) = 0.000335 + 0.325 + 0.607 + 0.0022 = **0.934535**

Σ [K(u) × Y_i] = -0.00077 + 0.1625 + 5.6451 + 0.0458 = **5.85263**

Step 3: Calculate Estimated Value

$$\hat{y}(1.3) = \frac{5.85263}{0.934535} = 6.262$$

Answer: ŷ(1.3) = 6.26 (rounded to 2 decimal places)

Question from Page 20, Question 3b (OR option) (Fifth Occurrence) [+5]

Question: You are given the following data:

x | 0.5 | 1.0 | 1.5 | 2.0
y | 1.2 | 2.3 | 3.1 | 3.8

Estimate the value of y at x = 1.2 using the Nadaraya-Watson estimator. Take bandwidth h = 0.3 and Gaussian kernel.

Solution:

Step 1: Calculate for x = 1.2, h = 0.3

Using Gaussian kernel K(u) = e^(-u²/2)

i	X_i	Y_i	$u = (1.2 - X_i)/0.3$	u^2	K(u) = e^(-u²/2)	K(u) × Y_i
1	0.5	1.2	(1.2-0.5)/0.3 = 0.7/0.3 = 2.333	5.444	e^(-5.444/2) = e^(-2.722) = 0.0657	0.0657 × 1.2 = 0.0788
2	1.0	2.3	(1.2-1.0)/0.3 = 0.2/0.3 = 0.667	0.445	e^(-0.445/2) = e^(-0.2225) = 0.801	0.801 × 2.3 = 1.8423
3	1.5	3.1	(1.2-1.5)/0.3 = -0.3/0.3 = -1.0	1.0	e^(-1/2) = e^(-0.5) = 0.607	0.607 × 3.1 = 1.8817
4	2.0	3.8	(1.2-2.0)/0.3 = -0.8/0.3 = -2.667	7.111	e^(-7.111/2) = e^(-3.5555) = 0.0286	0.0286 × 3.8 = 0.1087

Step 2: Sum the Components

Σ K(u) = 0.0657 + 0.801 + 0.607 + 0.0286 = **1.5023**

Σ [K(u) × Y_i] = 0.0788 + 1.8423 + 1.8817 + 0.1087 = **3.9115**

Step 3: Calculate Estimated Value

$$\hat{y}(1.2) = \frac{3.9115}{1.5023} = 2.604$$

Answer: ŷ(1.2) = 2.60 (rounded to 2 decimal places)

Question from Page 4, Question 4a (Sixth Occurrence) [+6] and Page 13, Question 4a (Seventh Occurrence) [+7]

These questions ask to **derive** the Nadaraya-Watson estimator rather than compute with data. Since they are theoretical derivations, I'll provide the derivation once.

Question: Derive the Nadaraya-Watson kernel regression estimator for the conditional mean E[Y|X=x]. Clearly show how the formula is obtained using kernel weighting.

Solution:

Step 1: Start with the Conditional Mean

We want to estimate: m(x) = E[Y | X = x]

Step 2: Use Joint Distribution Approach

$$m(x) = \frac{\int yf(x,y)dy}{\int f(x,y)dy} = \frac{\int yf(x,y)dy}{f(x)}$$

Where f(x,y) is the joint density and f(x) is the marginal density of X.

Step 3: Estimate Densities Using Kernel

Using kernel density estimation:

$$\hat{f}(x,y) = \frac{1}{n}\sum\nolimits_{i=1}^{n} K_h\,(x - X_i)K_h(y - Y_i)$$

$$\hat{f}(x) = \frac{1}{n}\sum\nolimits_{i=1}^{n} K_h\,(x - X_i)$$

Step 4: Substitute into the Ratio

$$\widehat{m}(x) = \frac{\int y \cdot \frac{1}{n}\sum_{i=1}^{n} K_h\,(x - X_i)K_h(y - Y_i)dy}{\frac{1}{n}\sum_{i=1}^{n} K_h\,(x - X_i)}$$

Step 5: Simplify the Numerator

$$\int y \cdot \frac{1}{n}\sum\nolimits_{i=1}^{n} K_h\,(x - X_i)K_h(y - Y_i)dy = \frac{1}{n}\sum\nolimits_{i=1}^{n} K_h\,(x - X_i)\int yK_h(y - Y_i)dy$$

Step 6: Evaluate the Inner Integral

For a kernel centered at Y_i, ∫ y K_h(y - Y_i) dy = Y_i (property of kernel)

Therefore:

$$\widehat{m}(x) = \frac{\frac{1}{n}\sum_{i=1}^{n} K_h\,(x - X_i)Y_i}{\frac{1}{n}\sum_{i=1}^{n} K_h\,(x - X_i)}$$

Step 7: Final Nadaraya-Watson Estimator

$$\widehat{m}(x) = \frac{\sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right)Y_i}{\sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right)}$$

This is a weighted average of the Y_i's, with weights proportional to the distance between x and X_i.

🥇 RANK 1: K-MEANS CLUSTERING [Asked 7 times]

Question from Page 4, Question 5b (First Occurrence) [+1]

Question: Apply the K-Means algorithm with K=2 on the following dataset: (160,55), (175,65), (172,60), (180,70), (178,68), (165,58). Use the first two data points as the initial centroids: (160,55) and (175,65). Perform the clustering for two iterations, and at each step:

- Assign points to the nearest centroid
- Recompute the centroids
- Show the cluster membership after each iteration

Solution:

Step 1: Initialization

- **Centroid C1** = (160, 55) [Point 1]
- **Centroid C2** = (175, 65) [Point 2]

Iteration 1:

Step 2: Calculate Distances to Centroids

Using Euclidean distance: $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

Point 1: (160, 55)

- Distance to C1: $\sqrt{[(160-160)^2 + (55-55)^2]} = \sqrt{[0 + 0]} = \mathbf{0}$
- Distance to C2: $\sqrt{[(160-175)^2 + (55-65)^2]} = \sqrt{[(-15)^2 + (-10)^2]} = \sqrt{[225 + 100]} = \sqrt{325} = \mathbf{18.03}$
- **Assign to Cluster 1** (closer to C1)

Point 2: (175, 65)

- Distance to C1: $\sqrt{[(175-160)^2 + (65-55)^2]} = \sqrt{[15^2 + 10^2]} = \sqrt{[225 + 100]} = \sqrt{325} = \mathbf{18.03}$
- Distance to C2: $\sqrt{[(175-175)^2 + (65-65)^2]} = \sqrt{[0 + 0]} = \mathbf{0}$
- **Assign to Cluster 2** (closer to C2)

Point 3: (172, 60)

- Distance to C1: $\sqrt{[(172-160)^2 + (60-55)^2]} = \sqrt{[12^2 + 5^2]} = \sqrt{[144 + 25]} = \sqrt{169} = \mathbf{13}$
- Distance to C2: $\sqrt{[(172-175)^2 + (60-65)^2]} = \sqrt{[(-3)^2 + (-5)^2]} = \sqrt{[9 + 25]} = \sqrt{34} = \mathbf{5.83}$

- **Assign to Cluster 2** (5.83 < 13)

Point 4: (180, 70)

- Distance to C1: $\sqrt{[(180-160)^2 + (70-55)^2]} = \sqrt{[20^2 + 15^2]} = \sqrt{[400 + 225]} = \sqrt{625} =$ **25**
- Distance to C2: $\sqrt{[(180-175)^2 + (70-65)^2]} = \sqrt{[5^2 + 5^2]} = \sqrt{[25 + 25]} = \sqrt{50} =$ **7.07**
- **Assign to Cluster 2** (7.07 < 25)

Point 5: (178, 68)

- Distance to C1: $\sqrt{[(178-160)^2 + (68-55)^2]} = \sqrt{[18^2 + 13^2]} = \sqrt{[324 + 169]} = \sqrt{493} =$ **22.20**
- Distance to C2: $\sqrt{[(178-175)^2 + (68-65)^2]} = \sqrt{[3^2 + 3^2]} = \sqrt{[9 + 9]} = \sqrt{18} =$ **4.24**
- **Assign to Cluster 2** (4.24 < 22.20)

Point 6: (165, 58)

- Distance to C1: $\sqrt{[(165-160)^2 + (58-55)^2]} = \sqrt{[5^2 + 3^2]} = \sqrt{[25 + 9]} = \sqrt{34} =$ **5.83**
- Distance to C2: $\sqrt{[(165-175)^2 + (58-65)^2]} = \sqrt{[(-10)^2 + (-7)^2]} = \sqrt{[100 + 49]} = \sqrt{149}$ = **12.21**
- **Assign to Cluster 1** (5.83 < 12.21)

Step 3: Iteration 1 - Cluster Membership

Point	**Coordinates**	**Assigned Cluster**
P1	(160, 55)	**Cluster 1**
P2	(175, 65)	**Cluster 2**
P3	(172, 60)	**Cluster 2**
P4	(180, 70)	**Cluster 2**
P5	(178, 68)	**Cluster 2**
P6	(165, 58)	**Cluster 1**

Cluster 1 Members: P1, P6
Cluster 2 Members: P2, P3, P4, P5

Step 4: Recompute Centroids

New Centroid C1 = Average of Cluster 1 points

- $\bar{x} = (160 + 165)/2 = 325/2 = \mathbf{162.5}$
- $\bar{y} = (55 + 58)/2 = 113/2 = \mathbf{56.5}$
- **C1 = (162.5, 56.5)**

New Centroid C2 = Average of Cluster 2 points

- $\bar{x} = (175 + 172 + 180 + 178)/4 = 705/4 = \mathbf{176.25}$
- $\bar{y} = (65 + 60 + 70 + 68)/4 = 263/4 = \mathbf{65.75}$
- **C2 = (176.25, 65.75)**

Iteration 2:

Step 5: Recalculate Distances to New Centroids

Point 1: (160, 55)

- Distance to C1: $\sqrt{[(160-162.5)^2 + (55-56.5)^2]} = \sqrt{[(-2.5)^2 + (-1.5)^2]} = \sqrt{[6.25 + 2.25]} = \sqrt{8.5} = \mathbf{2.92}$
- Distance to C2: $\sqrt{[(160-176.25)^2 + (55-65.75)^2]} = \sqrt{[(-16.25)^2 + (-10.75)^2]} = \sqrt{[264.06 + 115.56]} = \sqrt{379.62} = \mathbf{19.49}$
- **Assign to Cluster 1** (2.92 < 19.49)

Point 2: (175, 65)

- Distance to C1: $\sqrt{[(175-162.5)^2 + (65-56.5)^2]} = \sqrt{[12.5^2 + 8.5^2]} = \sqrt{[156.25 + 72.25]} = \sqrt{228.5} = \mathbf{15.12}$
- Distance to C2: $\sqrt{[(175-176.25)^2 + (65-65.75)^2]} = \sqrt{[(-1.25)^2 + (-0.75)^2]} = \sqrt{[1.56 + 0.56]} = \sqrt{2.12} = \mathbf{1.46}$
- **Assign to Cluster 2** (1.46 < 15.12)

Point 3: (172, 60)

- Distance to C1: $\sqrt{[(172-162.5)^2 + (60-56.5)^2]} = \sqrt{[9.5^2 + 3.5^2]} = \sqrt{[90.25 + 12.25]} = \sqrt{102.5} = \mathbf{10.12}$

- Distance to C2: $\sqrt{[(172-176.25)^2 + (60-65.75)^2]} = \sqrt{[(-4.25)^2 + (-5.75)^2]} = \sqrt{[18.06 + 33.06]} = \sqrt{51.12} = \mathbf{7.15}$
- **Assign to Cluster 2** (7.15 < 10.12)

Point 4: (180, 70)

- Distance to C1: $\sqrt{[(180-162.5)^2 + (70-56.5)^2]} = \sqrt{[17.5^2 + 13.5^2]} = \sqrt{[306.25 + 182.25]} = \sqrt{488.5} = \mathbf{22.10}$
- Distance to C2: $\sqrt{[(180-176.25)^2 + (70-65.75)^2]} = \sqrt{[3.75^2 + 4.25^2]} = \sqrt{[14.06 + 18.06]} = \sqrt{32.12} = \mathbf{5.67}$
- **Assign to Cluster 2** (5.67 < 22.10)

Point 5: (178, 68)

- Distance to C1: $\sqrt{[(178-162.5)^2 + (68-56.5)^2]} = \sqrt{[15.5^2 + 11.5^2]} = \sqrt{[240.25 + 132.25]} = \sqrt{372.5} = \mathbf{19.30}$
- Distance to C2: $\sqrt{[(178-176.25)^2 + (68-65.75)^2]} = \sqrt{[1.75^2 + 2.25^2]} = \sqrt{[3.06 + 5.06]} = \sqrt{8.12} = \mathbf{2.85}$
- **Assign to Cluster 2** (2.85 < 19.30)

Point 6: (165, 58)

- Distance to C1: $\sqrt{[(165-162.5)^2 + (58-56.5)^2]} = \sqrt{[2.5^2 + 1.5^2]} = \sqrt{[6.25 + 2.25]} = \sqrt{8.5} = \mathbf{2.92}$
- Distance to C2: $\sqrt{[(165-176.25)^2 + (58-65.75)^2]} = \sqrt{[(-11.25)^2 + (-7.75)^2]} = \sqrt{[126.56 + 60.06]} = \sqrt{186.62} = \mathbf{13.66}$
- **Assign to Cluster 1** (2.92 < 13.66)

Step 6: Iteration 2 - Cluster Membership

Point	Coordinates	Assigned Cluster
P1	(160, 55)	**Cluster 1**
P2	(175, 65)	**Cluster 2**
P3	(172, 60)	**Cluster 2**

Point	Coordinates	Assigned Cluster
P4	(180, 70)	**Cluster 2**
P5	(178, 68)	**Cluster 2**
P6	(165, 58)	**Cluster 1**

Cluster 1 Members: P1, P6
Cluster 2 Members: P2, P3, P4, P5

Step 7: Check for Convergence

Cluster membership is identical to Iteration 1 → Algorithm has converged.

Step 8: Final Centroids

- **Final Centroid C1** = (162.5, 56.5)
- **Final Centroid C2** = (176.25, 65.75)

Same Question from Page 8, Question 5b (Second Occurrence) [+2]

Same Question from Page 15, Question 5b (OR option) (Third Occurrence) [+3]

Same Question from Page 23, Question 9 (Fourth Occurrence) [+4]

Same Question from Page 5, Question 5b (Fifth Occurrence) [+5]

Same Question from Page 19, Question 5a (Sixth Occurrence) [+6]

(All identical to the first question - same solution as above)

Question from Page 5, Question 5b (Different Dataset) (Seventh Occurrence) [+7]

Question: What is clustering? Generate cluster from following dataset using K-means algorithm (take k = 2 and consider up to 2 iterations, use hamming distance).

T1: Bread, Jelly, Butter
T2: Bread, Butter
T3: Bread, Milk, Butter

T4: Coke, Bread
T5: Coke, Milk

Solution:

Step 1: Convert to Binary Format (Presence/Absence)

Item	Bread	Jelly	Butter	Milk	Coke
T1	1	1	1	0	0
T2	1	0	1	0	0
T3	1	0	1	1	0
T4	1	0	0	0	1
T5	0	0	0	1	1

Step 2: Initialization (Choose first two as centroids)

- **Centroid C1** = T1 = (1, 1, 1, 0, 0)
- **Centroid C2** = T2 = (1, 0, 1, 0, 0)

Iteration 1:

Step 3: Calculate Hamming Distances

Hamming distance = number of positions where bits differ

T1 to C1: (1,1,1,0,0) vs (1,1,1,0,0) → 0 differences → **Distance = 0**
T1 to C2: (1,1,1,0,0) vs (1,0,1,0,0) → Jelly differs → **Distance = 1**
→ Assign T1 to **Cluster 1**

T2 to C1: (1,0,1,0,0) vs (1,1,1,0,0) → Jelly differs → **Distance = 1**
T2 to C2: (1,0,1,0,0) vs (1,0,1,0,0) → 0 differences → **Distance = 0**
→ Assign T2 to **Cluster 2**

T3: (1,0,1,1,0)

- to C1: Compare (1,0,1,1,0) vs (1,1,1,0,0) → Jelly(1 vs 0), Milk(1 vs 0) → **2 differences → Distance = 2**
- to C2: Compare (1,0,1,1,0) vs (1,0,1,0,0) → Milk differs → **Distance = 1**
  → Assign T3 to **Cluster 2** (1 < 2)

T4: (1,0,0,0,1)

- to C1: Compare (1,0,0,0,1) vs (1,1,1,0,0) → Jelly(0 vs 1), Butter(0 vs 1), Coke(1 vs 0) → **3 differences** → **Distance = 3**
- to C2: Compare (1,0,0,0,1) vs (1,0,1,0,0) → Butter(0 vs 1), Coke(1 vs 0) → **2 differences → Distance = 2**
  → Assign T4 to **Cluster 2** (2 < 3)

T5: (0,0,0,1,1)

- to C1: Compare (0,0,0,1,1) vs (1,1,1,0,0) → All 5 positions differ? Let's check:
  Bread: 0 vs 1 ✓, Jelly: 0 vs 1 ✓, Butter: 0 vs 1 ✓, Milk: 1 vs 0 ✓, Coke: 1 vs 0 ✓ → **5 differences** → **Distance = 5**
- to C2: Compare (0,0,0,1,1) vs (1,0,1,0,0) →
  Bread: 0 vs 1 ✓, Jelly: 0 vs 0 ✗, Butter: 0 vs 1 ✓, Milk: 1 vs 0 ✓, Coke: 1 vs 0 ✓ → **4 differences** → **Distance = 4**
  → Assign T5 to **Cluster 2** (4 < 5)

Step 4: Iteration 1 - Cluster Membership

Transaction	Assigned Cluster
T1	**Cluster 1**
T2	**Cluster 2**
T3	**Cluster 2**
T4	**Cluster 2**
T5	**Cluster 2**

Cluster 1 Members: T1
Cluster 2 Members: T2, T3, T4, T5

Step 5: Recompute Centroids (Mode for each attribute)

New Centroid C1 = T1 = (1, 1, 1, 0, 0) [only one member]

New Centroid C2 = Mode of T2, T3, T4, T5

Attribute	T2	T3	T4	T5	Mode
Bread	1	1	1	0	**1** (appears 3 times)
Jelly	0	0	0	0	**0** (appears 4 times)
Butter	1	1	0	0	Mode tie (1 and 0 both appear twice) - choose 1? Usually choose most frequent or break tie
Milk	0	1	0	1	Mode tie (0 and 1 both appear twice)
Coke	0	0	1	1	Mode tie (0 and 1 both appear twice)

For tie-breaking, we need a rule. Let's use: if tie, keep previous value or use 0.

C2 = (1, 0, 1, 0, 0) (taking Butter=1, Milk=0, Coke=0 as tie-breaker)

Iteration 2:

Step 6: Recalculate Distances to New Centroids

C1 = (1, 1, 1, 0, 0)
C2 = (1, 0, 1, 0, 0)

T1: (1,1,1,0,0)

- to C1: 0 differences → **0**
- to C2: Compare (1,1,1,0,0) vs (1,0,1,0,0) → Jelly differs → **1**
  → Assign to **Cluster 1**

T2: (1,0,1,0,0)

- to C1: Compare (1,0,1,0,0) vs (1,1,1,0,0) → Jelly differs → **1**

- to C2: 0 differences → **0**
  → Assign to **Cluster 2**

T3: (1,0,1,1,0)

- to C1: Compare vs (1,1,1,0,0) → Jelly(0 vs 1), Milk(1 vs 0) → **2**
- to C2: Compare vs (1,0,1,0,0) → Milk differs → **1**
  → Assign to **Cluster 2**

T4: (1,0,0,0,1)

- to C1: Compare vs (1,1,1,0,0) → Jelly(0 vs 1), Butter(0 vs 1), Coke(1 vs 0) → **3**
- to C2: Compare vs (1,0,1,0,0) → Butter(0 vs 1), Coke(1 vs 0) → **2**
  → Assign to **Cluster 2**

T5: (0,0,0,1,1)

- to C1: Compare vs (1,1,1,0,0) → All 5 differ → **5**
- to C2: Compare vs (1,0,1,0,0) → Bread(0 vs 1), Butter(0 vs 1), Milk(1 vs 0), Coke(1 vs 0) → **4**
  → Assign to **Cluster 2**

Step 7: Iteration 2 - Cluster Membership

Transaction	**Assigned Cluster**
T1	**Cluster 1**
T2	**Cluster 2**
T3	**Cluster 2**
T4	**Cluster 2**
T5	**Cluster 2**

Same as Iteration 1 → Algorithm converged.

Step 8: Final Clusters

- **Cluster 1:** T1 (Bread, Jelly, Butter) - Transactions with Jelly

- **Cluster 2:** T2, T3, T4, T5 - Transactions without Jelly

🥈 RANK 4: PRINCIPAL COMPONENT ANALYSIS (PCA) [Asked 6 times]

Question from Page 4, Question 4b (First Occurrence) [+1]

Question: You are given the following two-dimensional data set consisting of 6 observations: (2,1), (3,5), (4,3), (5,6), (6,7), (7,8). Using the Principal Component Analysis (PCA) algorithm, compute the first principal component of the dataset.

Solution:

Step 1: Organize the Data

Observation	X	Y
1	2	1
2	3	5
3	4	3
4	5	6
5	6	7
6	7	8

n = 6

Step 2: Calculate Means

$\bar{x}$ = (2 + 3 + 4 + 5 + 6 + 7)/6 = 27/6 = **4.5**
$\bar{y}$ = (1 + 5 + 3 + 6 + 7 + 8)/6 = 30/6 = **5**

Step 3: Center the Data (Subtract Means)

Obs	$X - \bar{x}$	$Y - \bar{y}$
1	2 - 4.5 = -2.5	1 - 5 = -4
2	3 - 4.5 = -1.5	5 - 5 = 0
3	4 - 4.5 = -0.5	3 - 5 = -2
4	5 - 4.5 = 0.5	6 - 5 = 1
5	6 - 4.5 = 1.5	7 - 5 = 2
6	7 - 4.5 = 2.5	8 - 5 = 3

Step 4: Calculate Covariance Matrix

$Cov(X,X) = \Sigma(X - \bar{x})^2/(n-1) = [(-2.5)^2 + (-1.5)^2 + (-0.5)^2 + (0.5)^2 + (1.5)^2 + (2.5)^2]/5$
$= [6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25]/5 = 17.5/5 =$ **3.5**

$Cov(Y,Y) = \Sigma(Y - \bar{y})^2/(n-1) = [(-4)^2 + (0)^2 + (-2)^2 + (1)^2 + (2)^2 + (3)^2]/5$
$= [16 + 0 + 4 + 1 + 4 + 9]/5 = 34/5 =$ **6.8**

$Cov(X,Y) = \Sigma(X - \bar{x})(Y - \bar{y})/(n-1) = [(-2.5)(-4) + (-1.5)(0) + (-0.5)(-2) + (0.5)(1) + (1.5)(2) + (2.5)(3)]/5$
$= [10 + 0 + 1 + 0.5 + 3 + 7.5]/5 = 22/5 =$ **4.4**

Covariance Matrix =

$$\begin{bmatrix} 3.5 & 4.4 \\ 4.4 & 6.8 \end{bmatrix}$$

Step 5: Find Eigenvalues

Solve $|C - \lambda I| = 0$

$$\left| \begin{matrix} 3.5 - \lambda & 4.4 \\ 4.4 & 6.8 - \lambda \end{matrix} \right| = 0$$

$(3.5 - \lambda)(6.8 - \lambda) - (4.4)(4.4) = 0$

$23.8 - 3.5\lambda - 6.8\lambda + \lambda^2 - 19.36 = 0$

$\lambda^2 - 10.3\lambda + 4.44 = 0$

Using quadratic formula: $\lambda = [10.3 \pm \sqrt{(10.3^2 - 4\times1\times4.44)}]/2$
$\lambda = [10.3 \pm \sqrt{(106.09 - 17.76)}]/2$
$\lambda = [10.3 \pm \sqrt{88.33}]/2$
$\lambda = [10.3 \pm 9.40]/2$

$\lambda_1 = (10.3 + 9.40)/2 = 19.7/2 = \mathbf{9.85}$
$\lambda_2 = (10.3 - 9.40)/2 = 0.9/2 = \mathbf{0.45}$

Step 6: First Principal Component (Eigenvector for $\lambda_1 = 9.85$)

Solve $(C - \lambda_1 I)v = 0$:

$$\begin{bmatrix} 3.5 - 9.85 & 4.4 \\ 4.4 & 6.8 - 9.85 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -6.35 & 4.4 \\ 4.4 & -3.05 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From first equation: $-6.35v_1 + 4.4v_2 = 0 \rightarrow 4.4v_2 = 6.35v_1 \rightarrow v_2 = (6.35/4.4)v_1 = 1.443v_1$

Let $v_1 = 1$, then $v_2 = 1.443$

Normalize: $||v|| = \sqrt{(1^2 + 1.443^2)} = \sqrt{(1 + 2.082)} = \sqrt{3.082} = 1.755$

First Principal Component =

$$\begin{bmatrix} 1/1.755 \\ 1.443/1.755 \end{bmatrix} = \begin{bmatrix} 0.57 \\ 0.822 \end{bmatrix}$$

Step 7: Variance Explained

First PC explains: $\lambda_1/(\lambda_1 + \lambda_2) \times 100\% = 9.85/(9.85 + 0.45) \times 100\% = 9.85/10.3 \times 100\% = \mathbf{95.6\%}$

Answer: First Principal Component = $[0.57, 0.822]^T$, explaining 95.6% of variance

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Same Question from Page 7, Question 3b (Second Occurrence) [+2]

Same Question from Page 10, Question 4a (Third Occurrence) [+3]

Same Question from Page 18, Question 3b (Fourth Occurrence) [+4]

(All identical to the first question - same solution as above)

Question from Page 13, Question 5b (Fifth Occurrence) [+5]

Question: Given the covariance matrix:

$$\begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix}$$

i) Find the eigenvalues.
ii) Determine the proportion of total variance explained by the first principal component.

Solution:

Step 1: Find Eigenvalues

Solve $|C - \lambda I| = 0$

$$\begin{vmatrix} 5-\lambda & 2 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$(5 - \lambda)(3 - \lambda) - (2)(2) = 0$

$15 - 5\lambda - 3\lambda + \lambda^2 - 4 = 0$

$\lambda^2 - 8\lambda + 11 = 0$

Using quadratic formula: $\lambda = [8 \pm \sqrt{(64 - 44)}]/2 = [8 \pm \sqrt{20}]/2 = [8 \pm 4.472]/2$

$\lambda_1 = (8 + 4.472)/2 = 12.472/2 = \mathbf{6.236}$
$\lambda_2 = (8 - 4.472)/2 = 3.528/2 = \mathbf{1.764}$

Step 2: Variance Explained by First PC

Total variance = $\lambda_1 + \lambda_2 = 6.236 + 1.764 = 8$

Proportion = λ_1 / Total = 6.236/8 = **0.7795** = **77.95%**

Answer:

- Eigenvalues: $\lambda_1 = 6.236$, $\lambda_2 = 1.764$
- First PC explains 77.95% of total variance

Question from Page 10, Question 4a (Sixth Occurrence) [+6]

Question: Given data = {2,3,4,5,6,7,1,5,3,6,7,8}. Compute the principal component using PCA Algorithm.

(This is a univariate dataset - PCA on 1D data is trivial)

Solution:

Step 1: Organize Data

Data: 2, 3, 4, 5, 6, 7, 1, 5, 3, 6, 7, 8
n = 12

Step 2: Calculate Mean

$\Sigma x = 2+3+4+5+6+7+1+5+3+6+7+8 = 57$
$\bar{x} = 57/12 =$ **4.75**

Step 3: Calculate Variance

$\Sigma(x - \bar{x})^2 = (2-4.75)^2 + (3-4.75)^2 + ... + (8-4.75)^2$
$= (-2.75)^2 + (-1.75)^2 + (-0.75)^2 + (0.25)^2 + (1.25)^2 + (2.25)^2 + (-3.75)^2 + (0.25)^2 + (-1.75)^2 + (1.25)^2 + (2.25)^2 + (3.25)^2$
$= 7.5625 + 3.0625 + 0.5625 + 0.0625 + 1.5625 + 5.0625 + 14.0625 + 0.0625 + 3.0625 + 1.5625 + 5.0625 + 10.5625$
$= 52.25$

Variance = 52.25/(12-1) = 52.25/11 = **4.75**

Step 4: First Principal Component

For 1D data, the first principal component is just the data itself, with the eigenvector = 1.

The first PC explains 100% of the variance.

Answer: The first principal component is the data vector itself. Variance = 4.75.

🥉 RANK 5: MULTIPLE LINEAR REGRESSION [Asked 5 times]

Question from Page 3, Question 3a (First Occurrence) [+1]

Question: A real estate analyst collected data from 3 houses in a neighborhood to predict house prices based on two features: square footage (in 100s of square feet) and number of bedrooms.

Square footage (X1)	Number of bedrooms (X2)	House price (in $1000s) (Y)	
1	5	3	240
2	8	4	290
3	20	3	310

Find the regression equation: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

Solution:

Step 1: Organize Data

$n = 3$

X_1	X_2	Y
5	3	240
8	4	290
20	3	310

Step 2: Set up Matrices

$$X = \begin{bmatrix} 1 & 5 & 3 \\ 1 & 8 & 4 \\ 1 & 20 & 3 \end{bmatrix}, Y = \begin{bmatrix} 240 \\ 290 \\ 310 \end{bmatrix}, \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

Step 3: Compute X^TX

$$X^T X = \begin{bmatrix} 1 & 1 & 1 \\ 5 & 8 & 20 \\ 3 & 4 & 3 \end{bmatrix} \begin{bmatrix} 1 & 5 & 3 \\ 1 & 8 & 4 \\ 1 & 20 & 3 \end{bmatrix}$$

First row: [1×1 + 1×1 + 1×1, 1×5 + 1×8 + 1×20, 1×3 + 1×4 + 1×3] = [3, 33, 10]
Second row: [5×1 + 8×1 + 20×1, 5×5 + 8×8 + 20×20, 5×3 + 8×4 + 20×3] = [33, 25+64+400=489, 15+32+60=107]
Third row: [3×1 + 4×1 + 3×1, 3×5 + 4×8 + 3×20, 3×3 + 4×4 + 3×3] = [10, 15+32+60=107, 9+16+9=34]

$$X^T X = \begin{bmatrix} 3 & 33 & 10 \\ 33 & 489 & 107 \\ 10 & 107 & 34 \end{bmatrix}$$

Step 4: Compute X^TY

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 \\ 5 & 8 & 20 \\ 3 & 4 & 3 \end{bmatrix} \begin{bmatrix} 240 \\ 290 \\ 310 \end{bmatrix} = \begin{bmatrix} 240 + 290 + 310 \\ 5 \times 240 + 8 \times 290 + 20 \times 310 \\ 3 \times 240 + 4 \times 290 + 3 \times 310 \end{bmatrix}$$

$$= \begin{bmatrix} 840 \\ 1200 + 2320 + 6200 = 9720 \\ 720 + 1160 + 930 = 2810 \end{bmatrix}$$

Step 5: Solve $(X^TX)\beta = X^TY$

We need to find $(X^TX)^{-1}$. For a 3×3 matrix, we can solve using elimination or note that with n=3 and p=3, we have a perfect fit.

Let's solve the system:

$3\beta_0 + 33\beta_1 + 10\beta_2 = 840$ ...(1)
$33\beta_0 + 489\beta_1 + 107\beta_2 = 9720$ ...(2)
$10\beta_0 + 107\beta_1 + 34\beta_2 = 2810$ ...(3)

From (1): $\beta_0 = (840 - 33\beta_1 - 10\beta_2)/3$

Substitute into (3):
$10[(840 - 33\beta_1 - 10\beta_2)/3] + 107\beta_1 + 34\beta_2 = 2810$
$(8400 - 330\beta_1 - 100\beta_2)/3 + 107\beta_1 + 34\beta_2 = 2810$
$2800 - 110\beta_1 - 33.33\beta_2 + 107\beta_1 + 34\beta_2 = 2810$
$2800 - 3\beta_1 + 0.67\beta_2 = 2810$
$-3\beta_1 + 0.67\beta_2 = 10$ ...(4)

Substitute into (2):
$33[(840 - 33\beta_1 - 10\beta_2)/3] + 489\beta_1 + 107\beta_2 = 9720$
$11(840 - 33\beta_1 - 10\beta_2) + 489\beta_1 + 107\beta_2 = 9720$
$9240 - 363\beta_1 - 110\beta_2 + 489\beta_1 + 107\beta_2 = 9720$
$9240 + 126\beta_1 - 3\beta_2 = 9720$
$126\beta_1 - 3\beta_2 = 480$ ...(5)

Solve (4) and (5):
From (4): $0.67\beta_2 = 10 + 3\beta_1 \rightarrow \beta_2 = (10 + 3\beta_1)/0.67$

Substitute into (5):
$126\beta_1 - 3[(10 + 3\beta_1)/0.67] = 480$
$126\beta_1 - (30 + 9\beta_1)/0.67 = 480$
$126\beta_1 - 44.78 - 13.43\beta_1 = 480$
$112.57\beta_1 = 524.78$
$\beta_1 = 4.66$

Then $\beta_2 = (10 + 3\times4.66)/0.67 = (10 + 13.98)/0.67 = 23.98/0.67 = 35.79$

$\beta_0 = (840 - 33\times4.66 - 10\times35.79)/3 = (840 - 153.78 - 357.9)/3 = 328.32/3 = 109.44$

Step 6: Regression Equation

$Y = 109.44 + 4.66X_1 + 35.79X_2$

Question from Page 13, Question 4b (Second Occurrence) [+2]

Question: Fit multiple linear regression model.

Study time in hours	Score on math	Score on reading
2	70	65
4	80	75
6	90	85

Solution:

Step 1: Organize Data

Let X = Study time (hours)
Y_1 = Math score
Y_2 = Reading score

We need two separate regression equations.

For Math Score (Y_1):

n = 3

X	Y_1	X^2	XY_1
2	70	4	140
4	80	16	320
6	90	36	540
$\Sigma X=12$	$\Sigma Y_1=240$	$\Sigma X^2=56$	$\Sigma XY_1=1000$

$\bar{x} = 12/3 = 4$
$\bar{y}_1 = 240/3 = 80$

$\beta_1 = [\Sigma XY_1 - n(\bar{x})(\bar{y}_1)] / [\Sigma X^2 - n(\bar{x})^2] = [1000 - 3\times4\times80] / [56 - 3\times16] = [1000 - 960] / [56 - 48] = 40/8 = \mathbf{5}$

$\beta_0 = \bar{y}_1 - \beta_1\bar{x} = 80 - 5\times4 = 80 - 20 =$ **60**

Math Score Equation: $Y_1 = 60 + 5X$

For Reading Score (Y_2):

X	**Y_2**	**XY_2**
2	65	130
4	75	300
6	85	510
$\Sigma X=12$	$\Sigma Y_2=225$	$\Sigma XY_2=940$

$\bar{y}_2 = 225/3 = 75$

$\beta_1 = [\Sigma XY_2 - n(\bar{x})(\bar{y}_2)] / [\Sigma X^2 - n(\bar{x})^2] = [940 - 3\times4\times75] / [56 - 48] = [940 - 900] / 8 = 40/8 =$ **5**

$\beta_0 = \bar{y}_2 - \beta_1\bar{x} = 75 - 5\times4 = 75 - 20 =$ **55**

Reading Score Equation: $Y_2 = 55 + 5X$

Question from Page 19, Question 3b (Third Occurrence) [+3]

Question: You are given:

Obs	**X**	**Y**
1	1	3
2	2	5
3	3	7

Obs	X	Y
4	4	9
5	5	11

i) Estimate regression coefficients.
ii) Compute R^2 and Adjusted R^2
iii) Predict Y when X = 6.

Solution:

<u>Step 1: Calculate Components</u>

n = 5
$\Sigma X = 1+2+3+4+5 = 15$
$\Sigma Y = 3+5+7+9+11 = 35$
$\Sigma XY = (1\times3)+(2\times5)+(3\times7)+(4\times9)+(5\times11) = 3+10+21+36+55 = 125$
$\Sigma X^2 = 1+4+9+16+25 = 55$
$\Sigma Y^2 = 9+25+49+81+121 = 285$

$\bar{x} = 15/5 = 3$
$\bar{y} = 35/5 = 7$

<u>Step 2: Calculate Regression Coefficients</u>

$\beta_1 = [\Sigma XY - n(\bar{x})(\bar{y})] / [\Sigma X^2 - n(\bar{x})^2] = [125 - 5\times3\times7] / [55 - 5\times9] = [125 - 105] / [55 - 45] = 20/10 = \mathbf{2}$

$\beta_0 = \bar{y} - \beta_1\bar{x} = 7 - 2\times3 = 7 - 6 = \mathbf{1}$

Regression Equation: $\hat{Y} = 1 + 2X$

Step 3: Calculate SST, SSR, SSE

X	Y	$\hat{Y} = 1+2X$	$(Y - \bar{y})^2$	$(\hat{Y} - \bar{y})^2$	$(Y - \hat{Y})^2$
1	3	3	$(3-7)^2 = 16$	$(3-7)^2 = 16$	$(3-3)^2 = 0$
2	5	5	$(5-7)^2 = 4$	$(5-7)^2 = 4$	$(5-5)^2 = 0$

X	Y	$\hat{Y} = 1+2X$	$(Y - \bar{y})^2$	$(\hat{Y} - \bar{y})^2$	$(Y - \hat{Y})^2$
3	7	7	$(7-7)^2 = 0$	$(7-7)^2 = 0$	$(7-7)^2 = 0$
4	9	9	$(9-7)^2 = 4$	$(9-7)^2 = 4$	$(9-9)^2 = 0$
5	11	11	$(11-7)^2 = 16$	$(11-7)^2 = 16$	$(11-11)^2 = 0$
Σ			**Σ SST = 40**	**Σ SSR = 40**	**Σ SSE = 0**

$SST = \Sigma(Y - \bar{y})^2 = 40$
$SSR = \Sigma(\hat{Y} - \bar{y})^2 = 40$
$SSE = \Sigma(Y - \hat{Y})^2 = 0$

Step 4: Calculate R^2

$R^2 = SSR/SST = 40/40 =$ **1.00** (Perfect fit)

Step 5: Calculate Adjusted R^2

Adjusted $R^2 = 1 - [(1 - R^2)(n - 1)/(n - k - 1)]$

where k = number of predictors = 1
$n - k - 1 = 5 - 1 - 1 = 3$

Adjusted $R^2 = 1 - [(1 - 1)(4)/(3)] = 1 - [0 \times 4/3] =$ **1.00**

Step 6: Predict Y when X = 6

$\hat{Y} = 1 + 2(6) = 1 + 12 =$ **13**

Question from Page 20, Question 3b (Fourth Occurrence) [+4]

Question: Given dataset with 3 variables and 5 observations, explain how PCA is computed using eigenvalue decomposition. How do you determine variance explained?

(This is theoretical, not numerical)

Question from Page 23, Question 5 (Fifth Occurrence) [+5]

Question: Define Multivariate Linear Regression and formulate its equations in Matrix form. Write down the Assumptions that must hold before using model.

(This is theoretical, not numerical)

SHORT NOTES - TOP 10 MOST FREQUENT

1. Causation vs Correlation [Asked 8 times]

Definition:

- **Correlation** measures the strength and direction of a linear relationship between two variables
- **Causation** implies that one event is the result of the occurrence of the other event

Key Differences:

Aspect	Correlation	Causation
Meaning	Variables move together	One variable directly affects another
Direction	Symmetric (A↔B)	Directional (A→B)
Formula	$r = cov(X,Y)/(\sigma_x\sigma_y)$	No simple formula
Evidence	Statistical only	Requires controlled experiments

Examples:

- **Correlation only**: Ice cream sales and drowning deaths both increase in summer (correlated, but ice cream doesn't cause drowning)
- **Causation**: Smoking causes lung cancer (established through controlled studies)

Why Correlation ≠ Causation:

1. **Confounding variables**: Hidden third variable affects both
2. **Reverse causation**: B might cause A, not A causing B
3. **Spurious correlation**: Random chance

Tests for Causation:

- Randomized controlled trials
- Granger causality (time series)
- Natural experiments
- Hill's criteria

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2. Multicollinearity and VIF [Asked 5 times]

Definition:
Multicollinearity occurs when independent variables in a regression model are highly correlated with each other.

Problems Caused:

- Unstable coefficient estimates
- Inflated standard errors
- Difficulty determining individual variable importance
- Sensitive to small data changes

Detection - Variance Inflation Factor (VIF):

$$VIF_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is from regressing X_j on all other predictors

Interpretation:

- VIF = 1: No multicollinearity
- VIF between 1-5: Moderate correlation (acceptable)
- VIF > 5: High correlation (problematic)
- VIF > 10: Severe multicollinearity (must fix)

Remedies:

1. Remove one of the correlated variables
2. Combine variables (PCA, factor analysis)
3. Ridge regression
4. Collect more data

3. Logistic Regression [Asked 4 times]

Definition:
Logistic regression models the probability of a binary outcome using the logistic function.

Model:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k)}}$$

Or in logit form:

$$\log\left(\frac{p}{1 - p}\right) = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k$$

Key Features:

- Output is probability between 0 and 1
- Uses maximum likelihood estimation (not OLS)
- Decision boundary is linear in log-odds space

Difference from Linear Regression:

Aspect	Linear Regression	Logistic Regression
Outcome	Continuous	Binary/Categorical
Range	(-∞, ∞)	[0, 1]
Estimation	OLS	Maximum Likelihood
Assumptions	Normality, Homoscedasticity	No normality required
Evaluation	R^2, RMSE	Accuracy, AUC, Confusion Matrix

4. Statistical Significance and p-value [Asked 4 times]

Definition:

- **p-value**: Probability of obtaining results at least as extreme as observed, assuming null hypothesis is true
- **Statistical significance**: When p-value < significance level (α)

Interpretation:

- $p < 0.05$: Statistically significant (reject H_0)
- $p > 0.05$: Not statistically significant (fail to reject H_0)

Common Misconceptions:

- ❌ p-value is probability that H_0 is true
- ❌ p-value is probability that results occurred by chance
- ✅ p-value measures evidence against H_0

t-value:

$$t = \frac{\text{estimated coefficient}}{\text{standard error}}$$

Larger $|t|$ → smaller p-value → stronger evidence against H_0

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5. Zipf's Law [Asked 3 times]

Definition:
Zipf's law states that the frequency of any word is inversely proportional to its rank in the frequency table.

Formula:

$$f(k) \propto \frac{1}{k^s}$$

where $f(k)$ is frequency of the k-th ranked item, $s \approx 1$

Examples:

- Word frequencies: "the" appears twice as often as "of", which appears twice as often as "and"
- City populations: Largest city twice size of 2nd largest
- Website traffic: Top site gets twice traffic of 2nd ranked

Applications:

- Natural language processing

- Information retrieval
- Economics (income distribution)
- Urban planning

Limitations:

- Only applies to rank-frequency distributions
- s parameter varies by domain
- Doesn't hold for very small or very large ranks

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6. Permutation Tests and Partial Dependence Plots [Asked 3 times]

Permutation Tests (Variable Importance):

- Randomly shuffle values of one variable
- Measure drop in model performance
- Larger drop → more important variable

Algorithm:

1. Train model with all variables
2. For each variable:
    - Permute its values
    - Predict with permuted data
    - Calculate performance drop
3. Rank variables by importance

Partial Dependence Plots (PDP):
Shows marginal effect of a variable on predicted outcome.

Method:

1. Choose variable of interest
2. For each unique value:
    - Set all instances to that value
    - Average predictions

3. Plot value vs average prediction

Interpretation:

- Upward slope → Positive relationship
- Downward slope → Negative relationship
- Flat line → No relationship

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7. OLS Estimation [Asked 2 times]

Definition:
Ordinary Least Squares (OLS) estimates regression coefficients by minimizing sum of squared residuals.

Matrix Formulation:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Assumptions:

1. Linearity: $Y = X\beta + \varepsilon$
2. No perfect multicollinearity
3. Zero conditional mean: $E[\varepsilon|X] = 0$
4. Homoscedasticity: $Var(\varepsilon|X) = \sigma^2$
5. No autocorrelation
6. Normality of errors (for inference)

Properties (Gauss-Markov Theorem):

- Unbiased: $E[\hat{\beta}] = \beta$
- Best Linear Unbiased Estimator (BLUE)
- Minimum variance among linear unbiased estimators

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8. ARIMA Models [Asked 2 times]

Definition:
ARIMA (AutoRegressive Integrated Moving Average) models time series data.

Components:

- **AR(p)**: AutoRegressive - uses past values
$$Y_t = c + \phi_1 Y_{t-1} + \ldots + \phi_p Y_{t-p} + \varepsilon_t$$
- **I(d)**: Integrated - differencing to achieve stationarity
$$\Delta^d Y_t = \text{stationary series}$$
- **MA(q)**: Moving Average - uses past errors
$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q}$$

ARIMA(p,d,q):

- p = order of AR term
- d = number of differences
- q = order of MA term

ARMA vs ARIMA:

- ARMA: Stationary data (d=0)
- ARIMA: Non-stationary data (d>0)

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9. Granger Causality [Asked 2 times]

Definition:
A variable X Granger-causes Y if past values of X help predict Y beyond what past values of Y alone can predict.

Test Procedure:

1. Regress Y on past Y values (restricted model)
2. Regress Y on past Y and past X values (unrestricted model)
3. F-test to see if X coefficients are jointly significant

Limitations:

- Not true causality (only predictive causality)
- Requires stationary time series
- Sensitive to lag length selection
- Doesn't account for latent variables

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10. AIC and BIC [Asked 2 times]

Akaike Information Criterion (AIC):

$$AIC = -2\ln(L) + 2k$$

Bayesian Information Criterion (BIC):

$$BIC = -2\ln(L) + k\ln(n)$$

where:

- L = maximized likelihood
- k = number of parameters
- n = sample size

Interpretation:

- Lower values indicate better model
- Trade-off between fit and complexity
- BIC penalizes complexity more heavily than AIC

Selection Rule:

- Calculate AIC/BIC for all candidate models
- Choose model with minimum value
- Differences > 2 indicate significant preference

📊 SUMMARY OF SOLVED QUESTIONS

Rank	Topic	Frequency	Status
1	Pearson Correlation	7 times	✅ Solved (all 7 variations)
1	Nadaraya-Watson	7 times	✅ Solved (derivation + 5 computations)
1	K-Means Clustering	7 times	✅ Solved (numerical + Hamming distance)

Rank	Topic	Frequency	Status
4	PCA	6 times	✅ Solved (2D data + covariance matrix)
5	Multiple Linear Regression	5 times	✅ Solved (3 variations)
7.	Short Notes (Top 10)	2-8 times	✅ Provided

Total Numerical Questions Solved: 32 (covering all top frequency topics)